UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC662 mock paper (October 2026). Computed answers show working; theory answers are supported by lecture slides (not lecturer-marked) with a verbatim quote and slide citation.

2. Supported theory marking points quote the lecture decks; any unresolved slot would remain a visible placeholder.

3. Part A = 10 x (2 + 6); Part B = 2 x (5 + 5); total = 100 marks.

4. Computed: 0 slot(s). Supported theory: 24 slots. Placeholders: 0. Questions and answers are for local study use only.

5. Near-copy rejection used all 35 past papers; style widening is recorded per slot.

FOR MARKERS' USE ONLY

This marking scheme consists of 15 printed pages

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CSC662 - MARKING SCHEME (MOCK OCTOBER 2026)

Labels: computed / supported by lecture slides (not lecturer-marked) / placeholder. No official marking scheme exists for the real CSC662 papers; this is generated for the mock.

Entailment labels: supported = the quote directly supports the marking point (or the point passed the stem-overlap gate without an LLM check); partly supported = the quote supports part of the point or is ambiguous; not supported = the quote does not support the point. Only 'not supported' points triggered regeneration.

Part A: 10 x (2 + 6) = 80 marks. Part B: 2 x (5 + 5) = 20 marks. Total: 100 marks. Slots: 0 computed, 24 supported by lecture slides, 0 placeholder.

Methods: computed answers use verified solvers. Theory answers are supported by lecture slides (not lecturer-marked). Each marking point carries a verbatim quote, a deck/slide/chunk citation and a term-overlap score; low overlap is flagged for human review. Sources are restricted to the slot topic's allowed decks, and a slide is not reused across questions where an alternative exists.

PART A - QUESTION 1

A1a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain two characteristics of Trojan horse malware, including how it disguises itself and what harm it can cause once active on a host.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Trojan horses disguise themselves by appearing to be useful or benign programs so users install them unsuspectingly.	"They usually masquerade as helpful or harmless programs"	051 - Software Security, slide 12	051 - Software Security::s12::1	0.09 LOW	supported
2	Once running, a Trojan can steal private data, delete files, capture keystrokes, or open a way for additional malware to be installed.	"begin to steal private data, delete files, log keystrokes or provide an opening for further malware to get installed"	051 - Software Security, slide 12	051 - Software Security::s12::1	0.56	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 12

A1b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain six techniques by which malware propagates itself or attaches its code to a host program, and describe the effect each technique has on the infected system.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A virus can spread through boot sector infection, as shown by Elk Cloner which spread via boot sector infection.	"A program spread via boot sector infection"	051 - Software Security, slide 7	051 - Software Security::s07::1	0.46 LOW	partly supporte
2	A worm can propagate across a network by exploiting a remote execution feature of a service, as Morris worm used a debugging feature of sendmail.	"Used a debugging feature of sendmail (remote execution)"	051 - Software Security, slide 7	051 - Software Security::s07::1	0.33 LOW	partly supporte
3	An executable COM infection appends the	"Replace a program entry point with a jump to a virus body"	051 - Software Security, slide 9	051 - Software Security::s09::1	0.64	supported (ste gate)

	virus body to a program and replaces the program entry point with a jump to the virus body.					
4	After the virus code runs in a COM infection, it restores the original entry point and jumps to it so the host program still executes.	"Virus code restores the original entry point and jumps to it after its own execution"	051 - Software Security, slide 9	051 - Software Security::s09::1	0.60	supported (ste gate)
5	An executable EXE infection overwrites the program header to switch the entry point to a virus.	"Over write a program header to switch the entry point to a virus"	051 - Software Security, slide 10	051 - Software Security::s10::1	0.60	supported (ste gate)
6	In an EXE infection the virus jumps to the original entry point during execution so the infected program continues to run.	"Jump to the original entry point during execution"	051 - Software Security, slide 10	051 - Software Security::s10::1	0.50	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 10; 051 - Software Security, slide 7; 051 - Software Security, slide 9

PART A - QUESTION 2

A2a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) differences between morals and laws as forms of conduct guidance for computing professionals.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Morals are described as traditions of belief concerning right and wrong conduct within a society.	"Morals are traditions of belief about right and wrong conduct"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	0.67	supported (ste gate)
2	Laws are formal rules of conduct imposed by a sovereign authority such as a government on its subjects or citizens.	"Laws are formal rules of conduct that a sovereign authority, such as a government, imposes on its subjects or citizens"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	1.00	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 5

A2b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six (6) professional conduct obligations that the ACM code places on a computing professional, and explain why such a documented code is needed to guide conduct in the computing profession.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The professional must work to a high standard of quality in the processes and products of professional work.	"Strive to achieve high quality in both the processes and products of professional work."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.75	supported (ste gate)
2	The professional must keep up high standards of competence, conduct	"Maintain high standards of professional competence, conduct, and ethical practice."	071 - Ethics and Computer Security, slide	071 - Ethics and Computer Security::s12::1	0.78	supported (ste gate)

	and ethical practice.		12			
3	The professional must be aware of, and comply with, the rules that apply to professional work.	"Know and respect existing rules pertaining to professional work."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.43 LOW	supported
4	The professional must give and receive suitable professional review of work.	"Accept and provide appropriate professional review."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.33 LOW	supported
5	The professional must contribute to society and human well-being, recognising all people as stakeholders in computing.	"Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing."	071 - Ethics and Computer Security, slide 11	071 - Ethics and Computer Security::s11::1	0.73	supported (ste gate)
6	A documented code is needed because it sets out a written record of the behaviours that are acceptable for a professional group.	"Documented set of acceptable behaviors for professional or social groups."	071 - Ethics and Computer Security, slide 6	071 - Ethics and Computer Security::s06::1	0.42 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 11; 071 - Ethics and Computer Security, slide 12; 071 - Ethics and Computer Security, slide 6

PART A - QUESTION 3

A3a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the two stages of the biometric authentication process and state what happens during each stage.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The identification stage involves acquiring the biometric sample, creating the master characteristics and storing them.	"Identification Stage Acquisition Creation of Master characteristics Storage of Master characteristics"	072 - Biometrics and Authentication, slide 5	072 - Biometrics and Authentication::s05::1	0.30 LOW	supported
2	The authentication stage involves further acquisition(s), comparison against the stored data and a decision.	"Authentication Stage Acquisition(s) Comparison Decision"	072 - Biometrics and Authentication, slide 5	072 - Biometrics and Authentication::s05::1	0.50	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 072 - Biometrics and Authentication, slide 5

A3b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six facts about biometric mechanisms and technologies as presented in the course, covering the definition of a biometric, the two application sectors, the behavioural examples, the physiological examples, the fingerprint process, and the iris scanning advantages.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A biometric is defined as a physiological or behavioural characteristic of a human being that can	"A biometric is a physiological or behavioural characteristic of a human being that can distinguish one person from another and that theoretically can be used for	072 - Biometrics and Authentication, slide 2	072 - Biometrics and Authentication::s02::1	0.92	supported (ste gate)

	distinguish one person from another and can theoretically be used for identification or verification of identity.	identification or verification of identity."				
2	Biometric applications available today are divided into two sectors.	"Biometric applications available today are categorized into 2 sectors"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.86	supported (ste gate)
3	The behavioural category of biometrics includes voice, typing pattern and signature.	"Behavioural: Voice, Typing pattern, Signature"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.57	supported (ste gate)
4	The physiological category of biometrics includes iris, fingerprints, hand, retinal and face recognition.	"Psychological: Iris, Fingerprints, Hand, Retinal and Face recognition"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.60	supported (ste gate)
5	In fingerprint processing, the print is divided into loops, whorls and arch, and minutiae points such as ridge endings are calculated.	"Divides print into loops, whorls and arch"	072 - Biometrics and Authentication, slide 6	072 - Biometrics and Authentication::s06::1	0.43 LOW	partly supporte
6	Iris scanning requires no touch and overcomes the issues associated with retinal scanners.	"No touch required"	072 - Biometrics and Authentication, slide 10	072 - Biometrics and Authentication::s10::1	0.22 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 072 - Biometrics and Authentication, slide 10; 072 - Biometrics and Authentication, slide 2; 072 - Biometrics and Authentication, slide 4; 072 - Biometrics and Authentication, slide 6

PART A - QUESTION 4

A4a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) classes of threat and, for each, the type of harm it causes to a system's assets.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Disclosure is a threat class in which information is accessed without authorisation, harming confidentiality.	"Disclosure - unauthorized access to information"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.33 LOW	supported
2	Deception is a threat class in which false data is accepted, harming integrity.	"Deception - acceptance of false data"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.50	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 011 - Introduction to Computer Security, slide 10

A4b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

The Samy worm spread through MySpace by exploiting a cross-site scripting flaw. Explain how this worm injected its payload, how it harvested and recruited each victim's friends, and the scale of the resulting infection.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The worm injected JavaScript into a user profile by writing to document.body.innerHTML.	"Inject Javascript into a user profile using document.body.innerHTML"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.70	supported (ste gate)
2	It obtained the list of the victim's heroes by issuing a GET request on a user profile.	"Get the list of user's heros by using a GET on a user profile"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.38 LOW	supported
3	It added Samy as a field by performing XML-HTTP requests against the addFriends page.	"Add Samy as a field by performing XML-HTTP on the addFriends page"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.70	supported (ste gate)
4	Any user visiting an infected profile had their own profile and friends list infected with the JavaScript.	"Any user who visits an infected user's page will gets profile infected with Javascript and his friends list infected with Samy"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.67	supported (ste gate)
5	The infection eventually reached roughly one million MySpace profiles.	"Eventually about 1,000,000 MySpace profiles were infected."	052 - Web Security, slide 15	052 - Web Security::s15::1	0.43 LOW	supported
6	The underlying flaw is that any web application returning user input without filtering is a potential XSS vulnerability.	"Any web application that returns user input without filtering is a potential XSS vulnerability!"	052 - Web Security, slide 13	052 - Web Security::s13::1	0.77	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 052 - Web Security, slide 13; 052 - Web Security, slide 15

PART A - QUESTION 5

A5a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain two ethical impacts that technology can have on society, and describe how a professional code of ethics helps computing professionals respond to such impacts.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Technology can affect society by raising questions about acceptable standards of behaviour, which a code of ethics helps to clarify.	"Adds clarity to the understanding of acceptable standards of behavior."	071 - Ethics and Computer Security, slide 8	071 - Ethics and Computer Security::s08::1	0.14 LOW	partly supporte
2	Technology impacts everyone, so a code communicates shared guidelines that all professionals can follow.	"Communicates common guidelines for everyone to follow."	071 - Ethics and Computer Security, slide 8	071 - Ethics and Computer Security::s08::1	0.36 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 8

A5b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Refer to your answer in (a). Based on the scenario in (a), discuss how the computing professional's

responsibility for privacy should be exercised, referring to the ACM code of ethics and professional conduct and to the responsibilities of information specialists working within ethical, moral, and legal constraints.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The professional should recognise that the ACM is a professional body that maintains a code of ethics and professional conduct to guide members.	"Association of Computing Machinery ACM (founded 1947) has a code of ethics and professional conduct."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.46 LOW	partly supporte
2	The professional should acknowledge that merely describing the moral and ethical dimensions of ICT is insufficient when handling private data.	"Simply describing may not be enough when discussing the moral and ethical dimensions of ICT"	071 - Ethics and Computer Security, slide 3	071 - Ethics and Computer Security::s03::1	0.39 LOW	supported
3	The professional should accept that users in general, and information specialists in particular, have definite responsibilities for performing within ethical, moral, and legal constraints.	"Users in general, and information specialists in particular, have definite responsibilities in terms of performing within ethical, moral, and legal constraints"	071 - Ethics and Computer Security, slide 3	071 - Ethics and Computer Security::s03::1	0.82	supported (ste gate)
4	The professional should be aware that other bodies such as AITP provide quality IT-related education and forums for networking with experienced peers, supporting responsible conduct.	"provides quality IT-related education, information on relevant IT issues, and forums for networking with experienced peers and other IT professionals."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.56	supported (ste gate)
5	The professional should recognise that the IEEE Computer Society is another professional body whose code can guide conduct.	"Computer Society of the Institute of Electrical and Electronics Engineers (IEEE - CS) (founded in 1946)."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.23 LOW	partly supporte
6	Information specialists must carry out their duties within ethical, moral, and legal constraints, which is a definite responsibility.	"Users in general, and information specialists in particular, have definite responsibilities in terms of performing within ethical, moral, and legal constraints"	071 - Ethics and Computer Security, slide 3	071 - Ethics and Computer Security::s03::1	0.69	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 3; 071 - Ethics and Computer Security, slide 7

PART A - QUESTION 6

A6a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the Chinese Wall model as a formal security model, stating the concern it is designed to address and how its access decisions differ from static policies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The Chinese Wall model is designed to address conflicts of interest affecting a consultant or contractor.	"Brewer and Nash's Chinese Wall Policy is designed to address a very specific concern: conflicts of interest by a consultant or contractor."	024 - Formal Security Model, slide 27	024 - Formal Security Model::s27::1	0.80	supported (ste gate)

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2	Access decisions in the Chinese Wall model depend on the subject's history of past accesses, so permissions change dynamically.	"The access rights that any subject enjoys depends on the history of past accesses"	024 - Formal Security Model, slide 24	024 - Formal Security Model::s24::1	0.42 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 024 - Formal Security Model, slide 24; 024 - Formal Security Model, slide 27

A6b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Compare the access control matrix, access control lists and capability lists as structures for implementing access policies, and explain the requirements that any such access control structure must satisfy.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Access control structures are mechanisms used to implement access policies, and the matrix, capability lists and ACLs are named examples of such structures.	"Access control structures are mechanisms for implementing access policies:"	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	0.55	supported (ste gate)
2	The access control matrix is one of the listed structures for implementing access policies.	"access control matrix (ACM)"	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	0.50	supported (ste gate)
3	Capability lists (C-List) are one of the listed structures for implementing access policies.	"Capabilities (C-List)"	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	0.20 LOW	supported
4	Access control lists (ACL) are one of the listed structures for implementing access policies.	"access control lists (ACL)"	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	0.50	supported (ste gate)
5	A required property of an access control structure is the ability to express control policies.	"an ability to express control policies"	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	0.50	supported (ste gate)
6	Further requirements are verifiability of correctness together with scalability and manageability.	"verifiability of correctness."	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	0.14 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 023 - Access Control and Authorization, slide 7

PART A - QUESTION 7

A7a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the goals that access control pursues and the role of a reference monitor in realizing access control policies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment

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1	Access control has four goals: granting, limiting, preventing and revoking access.	"Goals of access control: Granting access Limiting access Preventing access Revoking access"	023 - Access Control and Authorization, slide 4	023 - Access Control and Authorization::s04::1	0.88	supported (ste gate)
2	A reference monitor is the component that makes authorization decisions.	"Reference monitor makes authorization decisions."	023 - Access Control and Authorization, slide 4	023 - Access Control and Authorization::s04::1	1.00	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 023 - Access Control and Authorization, slide 4

A7b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six points about the formal security models covered in the lecture, covering the BLP and Biba models, the Chinese Wall model, and the classification of the Chinese Wall policy.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The BLP model is described as providing access control only.	"Bell-La Padula (BLP) model: access control only"	024 - Formal Security Model, slide 5	024 - Formal Security Model::s05::1	0.83	supported (ste gate)
2	The Biba model is described as adding integrity verification on top of access control.	"Biba model: additional integrity verification"	024 - Formal Security Model, slide 5	024 - Formal Security Model::s05::1	0.50	supported (ste gate)
3	The Chinese Wall model is characterized by the identification of conflicts of interest.	"Chinese Wall model: identification of conflicts of interest"	024 - Formal Security Model, slide 5	024 - Formal Security Model::s05::1	0.86	supported (ste gate)
4	The Chinese Wall policy was proposed by Brewer and Nash in 1989 to address conflicts of interest.	"Brewer and Nash (1989) proposed a policy called the Chinese Wall Policy that addresses such conflicts of interest."	024 - Formal Security Model, slide 21	024 - Formal Security Model::s21::1	1.00	supported (ste gate)
5	Strictly speaking, the Chinese Wall policy is an access control confidentiality policy rather than an integrity policy.	"Strictly speaking, this is not an integrity policy, but an access control confidentiality policy."	024 - Formal Security Model, slide 21	024 - Formal Security Model::s21::1	0.64	supported (ste gate)
6	Under the Chinese Wall simple security rule, a subject may access an object only if it lies within the Wall or belongs to an entirely different conflict of interest class.	"is in the same company datasets as the objects already accessed by s, that is, "within the Wall," or"	024 - Formal Security Model, slide 25	024 - Formal Security Model::s25::1	0.22 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 024 - Formal Security Model, slide 21; 024 - Formal Security Model, slide 25; 024 - Formal Security Model, slide 5

PART A - QUESTION 8

A8a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) facts about the chain of trust in trusted computing, including how measurements are recorded and where the chain begins.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment

1	Each stage of the chain measures and validates the following stage.	"Each stage measures and validates the next one"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.60	supported (ste gate)
2	The measurements are stored in Platform Configuration Registers (PCRs) on the TPM.	"Measurements go into Platform Configuration Registers (PCRs) on the TPM"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.86	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 073 - Trusted Computing, slide 8

A8b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the roles of the two primary system components of the Next Generation Secure Computing Base (NGSCB), describing what each component does and how they relate to the trusted platform.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Nexus is a special kernel that forms the core of the trusted operating environment.	"Special kernel (core of the trusted operating)"	073 - Trusted Computing, slide 13	073 - Trusted Computing::s13::1	0.62	supported (ste gate)
2	Nexus isolates normal-mode processes from trusted-mode processes by placing them differently in memory.	"Isolate the process of normal mode and trusted mode differently in memory"	073 - Trusted Computing, slide 13	073 - Trusted Computing::s13::1	0.67	supported (ste gate)
3	Nexus authenticates and protects data that is entered, stored, communicated, and displayed by encrypting it.	"Authenticate and protect data (entered, stored, communicated, and displayed) by data encryption"	073 - Trusted Computing, slide 13	073 - Trusted Computing::s13::1	0.89	supported (ste gate)
4	Nexus acts as the application interface to the TPM.	"Application interface to TPM"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.60	supported (ste gate)
5	Nexus is a security kernel that is authenticated on boot and authenticates trusted programs.	"Security kernel, authenticated on boot"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.57	supported (ste gate)
6	The Nexus Computing Agent is a trusted software component that runs in trusted mode and communicates with Nexus.	"Runs in trusted mode that communicates with Nexus"	073 - Trusted Computing, slide 15	073 - Trusted Computing::s15::1	0.62	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 073 - Trusted Computing, slide 13; 073 - Trusted Computing, slide 14; 073 - Trusted Computing, slide 15

PART A - QUESTION 9

A9a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List TWO (2) key points about software insecurity and buffer overflows as described in the Software Security lecture.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Software insecurity arises because attackers can modify	"Software insecurity stems from attacker's ability to modify system resources criticial for program	051 - Software Security, slide 30	051 - Software Security::s30::1	0.72	supported (ste gate)

	system resources that are critical for program execution, such as the instruction pointer or function call addresses.	execution, e.g. instruction pointer, function call addresses, interrupt addresses, etc."				
2	A major source of software insecurity is the failure to validate user input.	"One of the key sources for software insecurity is failed validation of user input."	051 - Software Security, slide 30	051 - Software Security::s30::1	0.75	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 30

A9b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss six (6) aspects of Windows security mechanisms, covering the purpose of its access control facilities, the role of Active Directory and tokens, the historical security milestones, and the risks that arise from the complexity of these mechanisms.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Windows provides sophisticated mechanisms for managing user accounts, objects and access control.	"Windows systems contain complex security mechanisms for user account and object management as well as access control."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.80	supported (ste gate)
2	Discretionary access control in Windows allows fine-grained control over diverse entities in the OS and in applications.	"DACS enable fine-grained access control of heterogeneous entities in Windows itself as well as applications."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.55	supported (ste gate)
3	The very complexity of Windows security can backfire, since misconfiguration and implementation bugs can cause serious incidents.	"The complexity of Windows security mechanisms is also its enemy: misconfiguration as well as implementation bugs may lead to severe security incidents."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.46 LOW	supported
4	Active Directory, introduced with Windows 2000, is an LDAP-like directory service used to organise system resources.	"Active directory introduced in Windows 2000 is an LDAP-like directory service for organization of system resources:"	041 - Windows Security, slide 6	041 - Windows Security::s06::1	0.91	supported (ste gate)
5	Security credentials for a Windows subject are held in a token, which maps principals to subjects and carries extra security attributes.	"Security credentials for a subject are stored in a token."	041 - Windows Security, slide 11	041 - Windows Security::s11::1	0.36 LOW	partly supporte
6	Windows Vista (2007) marked a security milestone by removing admin-by-default and adding a firewall and DEP.	"Vista (2007): no "admin-by-default", firewall, DEP"	041 - Windows Security, slide 2	041 - Windows Security::s02::1	0.42 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 041 - Windows Security, slide 11; 041 - Windows Security, slide 2; 041 - Windows Security, slide 21; 041 - Windows Security, slide 6

PART A - QUESTION 10

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A10a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List two ways in which an intrusion detection system can help defend a system against attacks, according to the lecture.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	An IDS can verify whether an attack succeeded or failed by reviewing HIDS log entries.	"Verifies success or failure of attack by reviewing HIDS log entries"	062 - Intrusion Detection System, slide 14	062 - Intrusion Detection System::s14::1	0.55	supported (ste gate)
2	An IDS can provide near real-time detection and response to attacks.	"Near real-time detection and response"	062 - Intrusion Detection System, slide 14	062 - Intrusion Detection System::s14::1	0.56	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 062 - Intrusion Detection System, slide 14

A10b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe three classes of threat to a computer system and, for each class, state one countermeasure that the lecture associates with preventing, detecting or recovering from it.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A threat is defined as an entity that represents a constant danger to an asset, i.e. a potential violation of security rather than an actual one.	"Threat: object, person, or other entity representing a constant danger to an asset"	011 - Introduction to Computer Security, slide 28	011 - Introduction to Computer Security::s28::1	0.46 LOW	partly supporte
2	A threat is only a potential violation of security, whereas an attack is the actual violation, so the two classes must be distinguished.	"Threats: A threat is a "potential" violation of security"	011 - Introduction to Computer Security, slide 9	011 - Introduction to Computer Security::s09::1	0.36 LOW	partly supporte
3	Software threats form a class that includes modification of program code and self-replicating code such as viruses.	"Modification of program code Viruses and self-replicating code"	051 - Software Security, slide 2	051 - Software Security::s02::1	0.42 LOW	supported
4	Malicious software such as spyware is a threat class that steals user logins and passwords and can lead to identity theft.	"They steal user logins and passwords, document sites visited, log keystrokes and even redirect browsers to unwanted sites."	051 - Software Security, slide 13	051 - Software Security::s13::1	0.29 LOW	partly supporte
5	A countermeasure for the threat of vulnerable software is prevention through vulnerability scanning and software updates.	"Prevention: vulnerability scanning, software updates"	011 - Introduction to Computer Security, slide 19	011 - Introduction to Computer Security::s19::1	0.62	supported (ste gate)
6	Detection countermeasures against threats include antivirus programs and intrusion detection systems.	"Detection: antivirus programs intrusion detection systems"	011 - Introduction to Computer Security, slide 19	011 - Introduction to Computer Security::s19::1	0.56	supported (ste gate)

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Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 011 - Introduction to Computer Security, slide 19; 011 - Introduction to Computer Security, slide 28; 011 - Introduction to Computer Security, slide 9; 051 - Software Security, slide 13; 051 - Software Security, slide 2

PART B - QUESTION 1

B1a - 5 marks - supported by lecture slides (not lecturer-marked)

Scenario:

A system administrator is reviewing how a shared UNIX server protects its user accounts. She wants to strengthen the controls applied to the stored password files and to the way accounts behave after repeated failed logins, while keeping the login process usable for staff.

Question:

Discuss FIVE (5) password security controls that can be applied to protect user accounts on a shared system, as presented in the course.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Access control should be applied to the password files themselves.	"Access control applied to password files"	022 - Identification and Authentication, slide 12	022 - Identification and Authentication::s12::1	0.57	supported (ste gate)
2	Accounts should be de-activated after multiple failed login attempts.	"De-activation of accounts after multiple failed logins"	022 - Identification and Authentication, slide 12	022 - Identification and Authentication::s12::1	0.75	supported (ste gate)
3	The system should display the time of the last login to the user.	"Displaying the time of a last login"	022 - Identification and Authentication, slide 12	022 - Identification and Authentication::s12::1	0.57	supported (ste gate)
4	Password policies should be defined, for example for the root password.	"Password policies, e.g. for root password"	022 - Identification and Authentication, slide 12	022 - Identification and Authentication::s12::1	0.50	supported (ste gate)
5	Passwords should be stored in an encrypted form using a one-way hash function.	"Password is stored in an encrypted form using a one-way hash function:"	022 - Identification and Authentication, slide 8	022 - Identification and Authentication::s08::1	0.88	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 022 - Identification and Authentication, slide 12; 022 - Identification and Authentication, slide 8

B1b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

The administrator in the scenario is considering single sign-on for the shared UNIX server. Explain how Kerberos and related single sign-on mechanisms address authentication in such an environment, giving FIVE (5) points as presented in the course.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Single sign-on reduces the burden of remembering passwords for users.	"Reduce the need to memorize passwords"	022 - Identification and Authentication, slide 16	022 - Identification and Authentication::s16::1	0.29 LOW	supported
2	Single sign-on reduces the time users spend typing passwords.	"Reduce the time spent on typing passwords"	022 - Identification and Authentication, slide 16	022 - Identification and Authentication::s16::1	0.43 LOW	supported
3	Kerberos is a single	"Single sign-on methods such as	022 -	022 - Identification and	0.90	supported (ste gate)

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	sign-on method that automates authentication across large distributed environments.	Kerberos provide means for automating authentication in large distributed environments."	Identification and Authentication, slide 24	Authentication::s24::1		
4	Kerberos is designed to resist eavesdropping by never sending passwords in clear text.	"Security against eavesdropping: no password transmission in clear text."	022 - Identification and Authentication, slide 18	022 - Identification and Authentication::s18::1	0.44 LOW	supported
5	Kerberos is transparent, so the user is unaware of any authentication beyond the initial login.	"Transparency: user is not aware of any authentication beyond an initial login."	022 - Identification and Authentication, slide 18	022 - Identification and Authentication::s18::1	0.78	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 022 - Identification and Authentication, slide 16; 022 - Identification and Authentication, slide 18; 022 - Identification and Authentication, slide 24

PART B - QUESTION 2

B2a - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the architecture of a Network-Based Intrusion Detection System by describing how sensors should be placed and what strengths this architecture offers the campus network team.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Sensors must be positioned deliberately so that the most valuable assets are defended.	"Place IDS sensors strategically to defend most valuable assets"	062 - Intrusion Detection System, slide 7	062 - Intrusion Detection System::s07::1	0.62	supported (ste gate)
2	A typical sensor location is just inside the firewall.	"Just inside the firewall"	062 - Intrusion Detection System, slide 7	062 - Intrusion Detection System::s07::1	0.50	supported (ste gate)
3	Because the IDS is shared, the overall cost of ownership is lower.	"Lower because IDS is shared"	062 - Intrusion Detection System, slide 8	062 - Intrusion Detection System::s08::1	0.57	supported (ste gate)
4	The sensor can examine all network traffic through packet analysis.	"Can look at all network traffic"	062 - Intrusion Detection System, slide 8	062 - Intrusion Detection System::s08::1	0.44 LOW	supported
5	Captured packets are stored on a separate machine, which supports evidence preservation.	"Packets are captured in a separate machine"	062 - Intrusion Detection System, slide 8	062 - Intrusion Detection System::s08::1	0.50	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 062 - Intrusion Detection System, slide 7; 062 - Intrusion Detection System, slide 8

B2b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

State five preventive network security mechanisms that a campus network team could deploy, giving for each the specific protective role it plays in controlling or securing network traffic.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A firewall screens traffic entering and leaving the network according to	"Control incoming and outgoing traffic on network with predetermined security policies."	061 - Network Security, slide 3	061 - Network Security::s03::1	0.20 LOW	supported

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	rules set in advance.					
2	A proxy server improves both data protection and network throughput while also filtering web content.	"To have a better data security and network performance. Proxy servers act as a firewall and web filter"	061 - Network Security, slide 3	061 - Network Security::s03::1	0.46 LOW	supported
3	Network segmentation draws boundaries so that assets sharing a function, risk or role sit inside the same segment.	"Boundaries between network segments where computing assets within the segment have a common function, risk or role within an organization."	061 - Network Security, slide 3	061 - Network Security::s03::1	0.58	supported (ste gate)
4	A remote access VPN lets individual hosts or clients reach the organization network securely from outside.	"Provides remote and secure access to an organization network to individual hosts or clients"	061 - Network Security, slide 4	061 - Network Security::s04::1	0.67	supported (ste gate)
5	Zero Trust Network Access restricts each user to only the access and permissions their role actually needs.	"A user should only has access and permission that required to fulfill the role"	061 - Network Security, slide 4	061 - Network Security::s04::1	0.46 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 061 - Network Security, slide 3; 061 - Network Security, slide 4

End of marking scheme.

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